



[Figure 13-2](#): All $3!$ (that is, 6) permutations of three books on a bookshelf

To calculate this yourself, think about how you would come up with every permutation of n books. You have n possible choices for the first book; then $n - 1$ possible choices for the second book (that is, every book except the one you picked for the first book); then $n - 2$ possible choices for the third book; and so on. With 6 books, $6!$ results in $6 \times 5 \times 4 \times 3 \times 2 \times 1$, or 720 photos. Adding just one more book makes this $7!$, or 5,040 photos needed. Even for small n values, factorial time algorithms quickly become impossible to complete in a reasonable amount of time. If you had 20 books and could arrange them and take a photo every second, it would still take longer than the universe has existed to get through every possible permutation.

One well-known $O(n!)$ problem is the traveling salesperson conundrum. A salesperson must visit n cities and wants to calculate the distance travelled for all $n!$ possible orders in which they could visit them. From these calculations, they could determine the order that involves the shortest travel distance. In a region with many cities, this task proves impossible to complete in a timely way. Fortunately, optimized algorithms can find a short (but not guaranteed to be the shortest) route much faster than $O(n!)$.

Big O Measures the Worst-Case Scenario

Big O specifically measures the *worst-case scenario* for any task. For example, finding a particular book on an unorganized bookshelf requires you to start from one end and scan the books until you find it. You might get lucky, and the book you're looking for might be the first book you check. But you might be unlucky; it could be the last book you check or not on the bookshelf at all. So in a best-case scenario, it wouldn't matter if there were billions of books you had to search through, because you'll immediately find the one you're looking for. But that optimism isn't useful for algorithm analysis. Big O describes what happens in the unlucky case: if